

301 interactive simulations with real numerics — no install, no cloud, free forever.

The Wolfram Alpha of interactive simulation

Flagship Engines, Fields & GPU Compute ⁽¹⁹⁾

Visual Node Graph /studio/graph	Wire blocks into a live dataflow — the real editor. Symbolic calculus built in.
2D Fluid (CFD) /studio/fluid	Interactive incompressible Navier–Stokes via Stam's stable-fluids method.
Heat & Wave Fields /studio/fields	2D heat diffusion and the 1D wave equation, solved live.
AI Surrogate /studio/surrogate	Train an ML surrogate on our solver for instant predictions. The PhysicsX play.
GPU Compute (WebGPU) /studio/gpu	Hundreds of thousands of particles on your GPU via WGSL shaders.
WebGPU Fluid /studio/gpu-fluid	High-res GPU smoke advection you can stir.
GPU N-Body /studio/gpu-nbody	Thousands of bodies, full $O(n^2)$ gravity on the GPU.
GPU PDE Solver /studio/gpu-pde	Steady heat/Poisson via GPU Jacobi at high res.
GPU 3D Fluid /studio/gpu-fluid-3d	3D smoke, simulated + raymarched on the GPU.
Particle-Mesh N-Body /studio/gpu-nbody-pm	100k+ bodies via GPU particle-mesh gravity.
3D CFD /studio/cfd-3d	3D Navier–Stokes plume with z-slices.
3D N-Body /studio/3d	Gravitation in 3D with a drag-to-orbit camera.
3D FEA Space Frame /studio/fea-3d	3D structural analysis; orbit the deformed tower.

3D Heat Diffusion /studio/heat-3d	Volumetric heat with slices and orbit view.
FEA Truss /studio/fea	Finite-element structural analysis: forces & deflection.
Meshing + BCs /studio/mesh	Paint boundary conditions; solve steady heat.
Optimize + UQ /studio/optimize	Gradient descent and Monte-Carlo uncertainty.
RLC Circuit /studio/rlc	Step response and damping regimes.
Gradient Descent /studio/gradient-descent	Optimization on loss landscapes.

Physics, Mechanics & Thermodynamics ⁽²³⁾

Particle / N-Body /studio/particles	Gravity, orbits, and impulse collisions with a real symplectic integrator.
Electrostatics /studio/electromagnetics	Charges, potential heatmaps, and field lines.
Double Pendulum /studio/double-pendulum	The textbook chaotic system, integrated with RK4.
Projectile Motion /studio/projectile	Ballistics with air drag; tune angle and speed.
Ising Model /studio/ising	A live magnetic phase transition via Monte Carlo.
Strange Attractors /studio/attractors	Lorenz, Rössler, Aizawa & more in 3D.
Wave Interference /studio/wave-interference	Two-source ripple-tank fringes.
Random Walk /studio/random-walk	Brownian motion and diffusion.

Kepler Orbits /studio/kepler	Conic-section orbits under gravity.
Double-Slit /studio/double-slit	Quantum interference + diffraction.
Cloth / Spring-Mass /studio/cloth	Verlet cloth you can grab and swing.
Gravity Well /studio/gravity-well	Curved-spacetime rubber sheet + orbit.
Magnetic Pendulum /studio/magnetic-pendulum	Fractal basins of attraction.
Percolation /studio/percolation	A phase transition in connectivity.
DLA Growth /studio/dla	Branching fractals from random walks.
Ray Optics / Lenses /studio/ray-optics	Trace rays through lenses.
Gas in a Box /studio/gas	Kinetic theory → pressure ($PV=nRT$).
Elastic Collisions /studio/collisions	Momentum & energy conservation.
Buoyancy /studio/buoyancy	Archimedes — float or sink by density.
Blackbody Radiation /studio/blackbody	Planck curve + Wien's law.
Pendulum Wave /studio/pendulum-wave	Phase art from many periods.
Standing Waves /studio/standing-waves	Harmonics, nodes & antinodes.
Doppler Effect /studio/doppler	Moving-source wavefronts & sonic boom.

Waves, Optics & Acoustics ⁽¹⁴⁾

Snell's Law /studio/snells-law	Refraction & total internal reflection.
Diffraction Grating /studio/diffraction-grating	N-slit interference maxima.
Beats /studio/beats	Two close tones interfere.
Curved Mirrors /studio/mirror	Concave/convex ray tracing.
Prism Dispersion /studio/prism	White light into a spectrum.
Fourier Transform /studio/fft	Signal → frequency spectrum.
Harmonic Series /studio/harmonic-series	Overtones & timbre.
Equal Temperament /studio/equal-temperament	Tuning & cents.
Chladni Plates /studio/chladni	Nodal sand patterns.
Helmholtz Resonator /studio/helmholtz-resonator	Cavity resonance.
Room Modes /studio/room-modes	Standing waves in a room.
Reverb Time (Sabine) /studio/reverb-time	RT60 decay.
Sound Levels (dB) /studio/sound-levels	Decibel addition & falloff.
Additive Synthesis /studio/additive-synthesis	Tone from sine waves.

Photonics & Modern Optics ⁽⁸⁾

Laser Cavity /studio/laser-cavity	Gain, loss & threshold.
Fiber Optics /studio/fiber-optics	Total internal reflection.
Gaussian Beam /studio/gaussian-beam	Waist & divergence.
Polarization /studio/polarization	Malus's law.
Thin-Film Coating /studio/thin-film	Anti-reflection interference.
Single-Slit Diffraction /studio/single-slit	Fraunhofer pattern.
Bragg Mirror /studio/bragg-mirror	Dielectric reflector.
LED Band Gap /studio/led	Gap sets the color.

Quantum & Modern Physics ⁽⁸⁾

Particle in a Box /studio/particle-box	Quantized energy levels.
Quantum Tunneling /studio/quantum-tunneling	Through a forbidden barrier.
Bloch Sphere /studio/bloch-sphere	The state of a qubit.
Hydrogen Orbitals /studio/hydrogen-orbitals	Electron probability clouds.
Special Relativity /studio/special-relativity	Time dilation & contraction.
Photoelectric Effect /studio/photoelectric	Einstein's photon.

Quantum Harmonic Oscillator</studio/quantum-harmonic>

Evenly-spaced energy ladder.

Stern-Gerlach</studio/stern-gerlach>

Spin measurement.

Nuclear & Radiation (8)**Nuclear Binding Energy**</studio/binding-energy>

Fusion vs fission.

Fission Chain Reaction</studio/fission-reactor>Criticality factor k .**Fusion (Lawson)**</studio/fusion-lawson>

Triple product ignition.

Radiometric Dating</studio/radiometric-dating>

Isotope clock ages.

Radiation Shielding</studio/radiation-shielding>

Attenuation & HVL.

Radiation Dose</studio/radiation-dose>

Time, distance, shielding.

Reactor Kinetics</studio/reactor-kinetics>

Reactivity & period.

Neutron Transport</studio/neutron-transport>

Random walk & moderation.

Mathematics (22)**Dynamical Systems**</studio/dynamics>

Lorenz, SIR epidemics, pendulums, predator–prey, and reaction–diffusion.

Symbolic Math</studio/cas>

Parse, differentiate, simplify, solve, and plot — a real CAS in your browser.

Vector Fields</studio/vector-field>Plot any $F(x,y)$ as a live quiver diagram.

Notebook /studio/notebook	Cells of prose + symbolic math + compute.
Fractal Explorer /studio/fractals	Zoom into the Mandelbrot and Julia sets.
Fourier Series /studio/fourier	Build waves from sine harmonics.
Function Grapher /studio/grapher	Plot up to three functions at once.
3D Surface Plotter /studio/surface-3d	Orbit any $z = f(x, y)$ surface.
Matrix Calculator /studio/matrix	Multiply, invert, determinant, transpose.
Cellular Automata /studio/cellular-automata	Wolfram rules & Conway's Life.
Taylor Series /studio/taylor	Polynomial approximation, live.
Newton's Method /studio/newton	Root finding, tangent by tangent.
Distributions /studio/distributions	Normal, Poisson, binomial & more.
Complex Functions /studio/complex	Domain coloring of $f(z)$.
Lissajous Curves /studio/lissajous	Two sine waves, endless patterns.
Bifurcation Diagram /studio/bifurcation	Logistic map route to chaos.
Direction Field /studio/direction-field	Slope fields + solution curves.
Riemann Sums /studio/riemann	Integrals as stacked rectangles.

Eigenvectors /studio/eigenvectors	2×2 transforms & invariant directions.
Markov Chains /studio/markov	Converge to a stationary distribution.
Parametric Grapher /studio/parametric	Trace $x(t)$, $y(t)$ curves.
Euler vs RK4 /studio/numerical-methods	ODE integration accuracy.

Statistics & Data Science ⁽⁸⁾

Hypothesis Test /studio/hypothesis-test	p-values & rejection regions.
Linear Regression /studio/linear-regression	Least squares & R^2 .
Central Limit Theorem /studio/central-limit	Means go normal.
Confidence Intervals /studio/confidence-interval	What 95% really means.
Bayesian Inference /studio/bayes-inference	Prior × likelihood → posterior.
Bootstrap /studio/bootstrap	Resampling confidence.
PCA /studio/pca	Axes of maximum variance.
A/B Test /studio/ab-test	Two-proportion significance.

Probability & Games of Chance ⁽⁸⁾

Dice Probability /studio/dice-probability	Sum distributions.
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Poker Odds /studio/poker-odds	Five-card hand odds.
Roulette Edge /studio/roulette	House edge & EV.
Birthday Paradox /studio/birthday-paradox	23 people, even odds.
Gambler's Ruin /studio/gamblers-ruin	The house always wins.
Monty Hall /studio/monty-hall	Always switch.
Law of Large Numbers /studio/law-of-large-numbers	Averages converge.
Lottery Odds /studio/lottery-odds	Jackpot combinatorics.

Optimization & Operations Research ⁽⁸⁾

Linear Programming /studio/linear-programming	Optimize over constraints.
0/1 Knapsack /studio/knapsack	DP value maximization.
M/M/1 Queue /studio/queueing	Queueing theory.
Economic Order Quantity /studio/eoq	Inventory cost optimum.
Simulated Annealing /studio/simulated-annealing	Escape local minima.
Critical Path (CPM) /studio/critical-path	Project scheduling.
Game Theory (Nash) /studio/game-theory	2×2 equilibria.

Maximum Flow
</studio/max-flow>

Network flow & min cut.

Networks & Graph Theory ⁽⁸⁾

Shortest Path
</studio/shortest-path>

Dijkstra weighted graph.

Spanning Tree
</studio/spanning-tree>

Kruskal MST.

Graph Coloring
</studio/graph-coloring>

Chromatic number.

Small-World Network
</studio/small-world>

Watts-Strogatz rewiring.

PageRank
</studio/pagerank>

Importance by links.

Bipartite Matching
</studio/bipartite-matching>

Optimal assignment.

Network Centrality
</studio/centrality>

Who matters most.

Graph Traversal
</studio/graph-traversal>

BFS vs DFS.

Chemistry & Materials ⁽⁸⁾

Molecular Dynamics
</studio/molecular-dynamics>

Lennard-Jones atoms; melt a lattice.

Titration Curve
</studio/titration>

pH vs base added; equivalence point.

Reaction Kinetics
</studio/reaction-kinetics>

Rate laws & Arrhenius decay.

Chemical Equilibrium
</studio/equilibrium>

Le Chatelier shifts.

pH Calculator /studio/ph	pH/pOH and the acid-base scale.
Radioactive Decay /studio/radioactive-decay	Half-life exponential decay.
Maxwell-Boltzmann /studio/maxwell-boltzmann	Molecular speed distribution.
Ideal Gas Law /studio/ideal-gas	$PV = nRT$ with a live piston.

Materials & Thermodynamics ⁽⁸⁾

Stress-Strain Curve /studio/stress-strain	Tensile test to fracture.
Carnot Cycle /studio/carnot-cycle	Ideal engine efficiency.
Otto Cycle /studio/otto-cycle	Gasoline engine P-V.
Phase Diagram /studio/phase-diagram	Solid, liquid, gas.
Thermal Expansion /studio/thermal-expansion	Growth with temperature.
Fatigue (S-N) /studio/fatigue	Cyclic life to failure.
Entropy /studio/entropy	The second law in action.
Thermal Resistance /studio/thermal-resistance	R-value composite wall.

Computer Science & AI ⁽¹⁰⁾

Sorting Visualizer /studio/sorting	Watch algorithms sort in real time.
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Boids Flocking /studio/boids	Emergent flocking from three rules.
A* Pathfinding /studio/pathfinding	Shortest path, draw walls.
Maze Generator /studio/maze	Carve & solve perfect mazes.
Neural Network /studio/neural-net	Train an MLP, watch it learn.
k-Means Clustering /studio/kmeans	Unsupervised clustering, live.
Convex Hull /studio/convex-hull	Wrap points in $O(n \log n)$.
L-System Fractals /studio/l-system	Grow plants from grammars.
Turing Machine /studio/turing-machine	The model of computation.
Image Convolution /studio/convolution	Kernels behind every CNN.

Cryptography & Information (8)

RSA Encryption /studio/rsa	Public-key crypto.
Diffie-Hellman /studio/diffie-hellman	Shared-secret exchange.
Classical Ciphers /studio/classical-ciphers	Caesar & Vigenère.
Hash Avalanche /studio/hash-avalanche	One bit changes all.
Shannon Entropy /studio/shannon-entropy	Info & password strength.

Huffman Coding /studio/huffman	Optimal compression.
Hamming Code /studio/hamming-code	Single-error correction.
Elliptic Curve /studio/elliptic-curve	ECC point addition.

Complex Systems & Chaos ⁽⁹⁾

Traffic Flow /studio/traffic	Phantom jams from random braking.
Forest Fire Model /studio/forest-fire	Self-organized criticality.
Abelian Sandpile /studio/sandpile	Avalanches build a fractal.
Schelling Segregation /studio/schelling	Mild bias, sharp segregation.
Langton's Ant /studio/langtons-ant	Two rules, emergent highway.
Reaction-Diffusion /studio/reaction-diffusion	Gray-Scott Turing patterns.
Elementary CA /studio/wolfram-ca	Wolfram's 256 rules.
Genetic Algorithm /studio/genetic-algorithm	Evolve toward the peak.
Ant Colony Optimization /studio/ant-colony	Swarm-solved TSP.

Biology & Medicine ⁽¹⁰⁾

Epidemic Network /studio/epidemic-network	Agent-based SIR on a contact graph.
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Spatial Predator-Prey /studio/predator-prey	Agent-based ecology with waves.
SIR Epidemic /studio/sir-model	Outbreak compartment model.
Hodgkin-Huxley Neuron /studio/neuron	Action potential spikes.
Lotka-Volterra /studio/lotka-volterra	Predator-prey cycles.
Hardy-Weinberg /studio/hardy-weinberg	Allele frequency & selection.
Enzyme Kinetics /studio/enzyme-kinetics	Michaelis-Menten curves.
Logistic Growth /studio/logistic-growth	Population carrying capacity.
Genetic Drift /studio/genetic-drift	Wright-Fisher allele drift.
Sequence Alignment /studio/sequence-alignment	Needleman-Wunsch DNA.

Astronomy & Space (8)

Exoplanet Transit /studio/exoplanet-transit	Light-curve transit method.
H-R Diagram /studio/hr-diagram	The map of the stars.
Lagrange Points /studio/lagrange-points	Restricted 3-body equilibria.
Roche Limit /studio/roche-limit	When tides shred a moon.
Hubble's Law /studio/hubble-law	The expanding universe.

Telescope Optics /studio/telescope	Resolution & magnification.
Stellar Parallax /studio/parallax	Distance from angle shift.
Escape Velocity /studio/escape-velocity	Speed to leave a world.

Aerospace & Flight (10)

Rocket Equation /studio/rocket	Tsiolkovsky Δv budget calculator.
Orbital Transfer /studio/orbital-transfer	Hohmann transfer Δv .
Airfoil Lift & Drag /studio/airfoil-polar	Wing polar & stall.
Glide Performance /studio/glide	Range from altitude.
Rocket Staging /studio/rocket-staging	Multi-stage delta-v.
Atmospheric Reentry /studio/reentry	Heating & g-load.
Standard Atmosphere /studio/standard-atmosphere	ISA T/P/density.
Mach Cone /studio/mach-cone	Supersonic shock.
Propeller Thrust /studio/propeller	Actuator-disk theory.
Orbital Elements /studio/orbital-elements	Shape of an orbit.

Earth, Climate & Environment ⁽⁸⁾

Energy Balance /studio/energy-balance	Greenhouse equilibrium temp.
Daisyworld /studio/daisyworld	Planetary self-regulation.
Milankovitch Cycles /studio/milankovitch	Orbital forcing of ice ages.
Tsunami Propagation /studio/tsunami	Shallow-water wave shoaling.
Carbon Cycle /studio/carbon-cycle	Reservoir box model.
Earthquake Shaking /studio/seismic	Magnitude, energy, intensity.
Groundwater Drawdown /studio/groundwater	Well cone of depression.
Lapse Rate /studio/lapse-rate	Atmosphere & cloud base.

Meteorology & Weather ⁽⁸⁾

Atmospheric Stability /studio/atmospheric-stability	CAPE & convection.
Coriolis Effect /studio/coriolis	Why winds curve.
Hurricane Model /studio/hurricane	Pressure-driven winds.
Wind Chill & Heat Index /studio/wind-chill	Apparent temperature.
Psychrometrics /studio/psychrometrics	Dew point & wet bulb.
Geostrophic Wind /studio/geostrophic-wind	Wind along isobars.

Rosby Waves
</studio/rossby-waves>

Meandering jet stream.

Tornado Vortex
</studio/rankine-vortex>

Rankine wind profile.

Environmental & Energy ⁽⁸⁾

Solar PV System
</studio/solar-panel>

Panels to kilowatt-hours.

Wind Turbine Power
</studio/wind-power>

Betz-limited power curve.

Hydroelectric Power
</studio/hydro-power>

Head × flow × gravity.

Heat Pump COP
</studio/heat-pump>

Heat moved vs energy in.

EV Range & Efficiency
</studio/ev-efficiency>

Speed vs range.

Battery Storage
</studio/battery-storage>

Usable capacity & backup.

Carbon Footprint
</studio/carbon-footprint>

Annual CO₂ breakdown.

Levelized Cost of Energy
</studio/lcoe>

True \$/MWh.

Finance & Quantitative ⁽⁸⁾

Black-Scholes
</studio/black-scholes>

Option pricing + Greeks.

Option Payoff
</studio/option-payoff>

Strategy P/L diagrams.

Monte Carlo (GBM)
</studio/monte-carlo>

Price-path distributions.

Efficient Frontier /studio/efficient-frontier	Portfolio risk vs return.
Value at Risk /studio/value-at-risk	VaR & expected shortfall.
Bond Pricing /studio/bond-pricing	Price, yield & duration.
Compound Interest /studio/compound-interest	Investment growth over time.
Loan Amortization /studio/amortization	Mortgage payment breakdown.

Economics & Markets ⁽⁸⁾

Supply & Demand /studio/supply-demand	Market equilibrium.
Price Elasticity /studio/elasticity	Demand sensitivity.
Monopoly Pricing /studio/monopoly	Market power & DWL.
Cobb-Douglas /studio/cobb-douglas	Optimal input mix.
Lorenz & Gini /studio/lorenz-gini	Income inequality.
Laffer Curve /studio/laffer-curve	Tax rate vs revenue.
Comparative Advantage /studio/comparative-advantage	Gains from trade.
Consumer Choice /studio/indifference-curves	Utility maximization.

Structural & Civil Engineering ⁽⁸⁾

Beam Deflection /studio/beam-deflection	Sag & bending moment.
Column Buckling /studio/column-buckling	Euler critical load.
Mohr's Circle /studio/mohrs-circle	2D stress transformation.
Shear & Moment /studio/shear-moment	SFD/BMD diagrams.
Retaining Wall /studio/retaining-wall	Rankine earth pressure.
Seismic Base Shear /studio/base-shear	Equivalent lateral force.
Soil Bearing /studio/soil-bearing	Terzaghi capacity.
Concrete Beam /studio/concrete-beam	RC flexural capacity.

Electrical & Signal Processing ⁽⁸⁾

Bode Plot /studio/bode-plot	Frequency response.
Filter Designer /studio/filter-designer	RC/RLC filter response.
PID Controller /studio/pid-control	Step-response tuning.
Transmission Line /studio/transmission-line	VSWR & reflections.
Op-Amp Circuits /studio/op-amp	Amplifier configurations.
Three-Phase Power /studio/three-phase	Grid power & phasors.

Transistor Bias
</studio/transistor-bias>

BJT Q-point load line.

Sampling & Aliasing
</studio/aliasing>

The Nyquist theorem.

Robotics, Control & Kinematics ⁽⁸⁾

Forward Kinematics
</studio/forward-kinematics>

Joint angles → end effector.

Inverse Kinematics
</studio/inverse-kinematics>

Target → joint angles.

Differential Drive
</studio/differential-drive>

Wheel speeds → path.

Cart-Pole Balance
</studio/cart-pole>

Inverted pendulum control.

Quadcopter Control
</studio/quadcopter>

Nested PID flight.

RRT Path Planning
</studio/rrt>

Sampling-based planner.

DC Motor
</studio/dc-motor>

Speed-torque & step.

Kalman Filter
</studio/kalman-filter>

Optimal state estimation.

Manufacturing & Process ⁽⁸⁾

Tolerance Stack-Up
</studio/tolerance-stackup>

Worst-case vs RSS.

Process Capability
</studio/process-capability>

Cp / Cpk & defects.

OEE
</studio/oee>

Equipment effectiveness.

CNC Feeds & Speeds /studio/cnc-feeds-speeds	Cutting parameters.
Line Balancing /studio/line-balancing	Takt time & stations.
SPC Control Chart /studio/control-chart	In control or not.
Learning Curve /studio/learning-curve	Cost falls with volume.
Little's Law /studio/littles-law	$WIP = \text{throughput} \times \text{time}$.

Geospatial & Navigation ⁽⁸⁾

Great-Circle Distance /studio/haversine	Haversine on a sphere.
Map Projection /studio/map-projection	Mercator distortion.
GPS Trilateration /studio/gps-trilateration	Position from distances.
Viewshed /studio/viewshed	Line-of-sight visibility.
Spatial Interpolation /studio/spatial-interpolation	IDW surface from samples.
Dead Reckoning /studio/dead-reckoning	Navigation with drift.
Rhumb Line /studio/rhumb-line	vs great circle.
Point in Polygon /studio/point-in-polygon	Ray-casting geofence.

First Responders (Fire | Police | EMS | Hazmat) ⁽⁸⁾

Wildfire Spread /studio/fire-spread	Wind + slope fire behavior.
Hazmat Plume /studio/hazmat-plume	Gaussian dispersion + PAD.
Building Evacuation /studio/evacuation	Egress flow & clear time.
START Triage /studio/triage	Mass-casualty triage tool.
Fire Hose Hydraulics /studio/hose-flow	Friction loss & pump pressure.
Skid-to-Stop Speed /studio/skid-to-stop	Accident reconstruction.
Radio Range /studio/radio-range	VHF/UHF link budget.
Blast Standoff /studio/blast-standoff	Overpressure & safe distance.

Sports & Games Analytics ⁽⁸⁾

Elo Rating /studio/elo-rating	Win probability & updates.
Pythagorean Expectation /studio/pythagorean-expectation	Wins from points.
Marathon Pacing /studio/marathon-pacing	Splits & strategy.
Magnus Effect /studio/magnus-effect	Spinning ball curve.
Expected Goals (xG) /studio/xg-model	Shot quality model.
Win Probability /studio/win-probability	Live game odds.

Tournament Odds</studio/tournament-bracket>

Bracket simulation.

Basketball Shot Arc</studio/shot-arc>

Optimal release angle.